

Zaid Duraid

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EDUCATION

Memorial University of Newfoundland

St. John's, NL, Canada

Bachelor of Engineering: **Computer Engineering** - CGPA: 4.00/4.00

Sep. 2022 - Apr. 2027

- **Dean's List (2022-2026)**
- **Scholarships:** Memorial University Alumni Entrance, Fry Family Foundation Dean's Leadership, Innovasea Computer Engineering, Verafin Inc. Computer Engineering, Charlie Sheppard Memorial - Hatch

EXPERIENCE

Firmware Validation Engineer Intern

Jan. 2026 - Apr. 2026, Aug. 2026 - Present

Tesla Inc.

Palo Alto, California, USA

- Implemented **LIN in Verilog on a Xilinx FPGA** for processor-in-the-loop (PIL) validation, with a Linux driver and fault injection paths, helping unblock PIL coverage for ~92% of drive inverter (DI) firmware configs
- Migrated a novel-design **Simulink PMSR motor model to a Xilinx FPGA in Verilog**, enabling developers to seamlessly conduct PIL dyno testing in realtime for DI firmware configs not ready for high-load HIL
- Developed **PIL fault-handling validation tests in Pytest** for verifying correct and timely fault responses across DI firmware configs, including 6+ fault scenarios across gate driver, power, and electrical oil pump systems
- Unblocked PIL for 50% of brake-by-wire firmware configs via **Verilog BLDC motor & SPI gate-driver models**

Advanced Research Electrical Engineer Intern

May 2025 - Aug. 2025

Solace Power

St. John's, NL, Canada

- Improved the design of a multi-MHz synchronous rectifier by adding a **PLL for closed-loop control** to maintain operation during ~100 μ s loss of switch gate driver signal
- Designed, built, and tested ~10 **single-layer PCBs using KiCad and in-house tools** to rapidly verify new subcircuits designs (e.g., RC inverter delay line for phase shift on GaN FET driver signal) into multilayer PCBs
- Researched **EMC solutions for a tens of MHz wireless power system**, reducing noise on key harmonics
- Wrote **C firmware for a TI C2000 microcontroller** to generate internal-clock or interrupt-driven PWM signals up to 9 MHz, with adjustable frequency, duty cycle, and phase shift via CANBUS

FPGA Hardware Engineer Intern

Jan. 2024 - Apr. 2024, Sep. 2024 - Dec. 2024

Avalon Holographics Inc.

St. John's, NL, Canada

- Led the development of new **internal verification tools using Python** that integrated with existing testbenches to verify hardware image correction features for the holographic display and reduce iteration time by 10x
- Developed and documented **simulation testbenches for FPGA designs in SystemVerilog** that integrate with internal verification tools, reducing critical bug discovery from ~2 months on hardware to ~2-3 days in simulation
- Followed internal test processes by **assembling hardware, executing tests, and running TCL scripts** for the bring-up of new/refurbished PCBs and validation of FPGA bitstream releases

PROJECTS & EXTRACURRICULARS

Part Time Founding Engineer

Dec. 2025 - July 2026

OmaScan

TypeScript, Node.js, React, React Three Fiber, AWS CDK

- Contributed to a full-stack cloud application for uploading, hosting, and interacting with 3D GLB scans for occupational therapist (OT) home assessments, **hosting real patient data for dozens of pilot OTs**
- Integrated PIPEDA-compliant product analytics using PostHog, **providing previously unavailable visibility into daily active users and in-app usage patterns** across development, staging, and production environments

Student Design Team Software Co-Lead

Aug. 2023 - Dec. 2025

Eastern Edge Robotics

Python, C/C++

- Built **ROS2 applications on a Raspberry Pi** to interface with ROV hardware over UART and I2C
- Implemented a **Gazebo ROV simulation environment** with C++ plugins for validating the full software stack
- Competed and presented alongside the team at the International MATE ROV Competition Explorer Class in 2023, 2024, and 2025, **placing 11th, 9th, and 3rd in the world out of around 20-30 teams**, respectively

SKILLS

Software/Firmware/HDL: Verilog/SystemVerilog, C/C++, Python, Golang, Rust, TypeScript

Hardware: Vivado, Quartus, Questa, KiCad, LTspice, Altium

Tools: Git, Claude Code, Docker, MATLAB/Simulink, FreeRTOS, ROS2, Gazebo, Solidworks, Onshape